

# Operations Research II: Algorithms

## Gradient Descent and Newton's Method

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# Road map

- ▶ **Introduction.**
- ▶ Gradient descent.
- ▶ Newton's method.

# Algorithms for nonlinear programming

- ▶ In many cases, we need to solve NLPs.
  - ▶ We rely on **numerical algorithms** for obtaining a numerical solution.
  - ▶ Typically the focus on an engineering application.
- ▶ To apply an algorithm, we need to first get the values of all parameters.
- ▶ Many NLP algorithms run in the following way:
  - ▶ **Iterative**: The algorithm moves to a point in one iteration, and then starts the next iteration starting from this point.
  - ▶ **Repetitive**: In each iteration, it repeats some steps.
  - ▶ **Greedy**: In each iteration, it seeks for some “best” thing achievable in that iteration.
  - ▶ **Approximation**: Relying on first-order or second-order approximation of the original program.

# Limitations of NLP algorithms

- ▶ NLP algorithms certainly have their limitations.
- ▶ It may **fail to converge**.
  - ▶ An algorithm converges to a solution if further iterations do not modify the current solution “a lot.”
  - ▶ Sometimes an algorithm may fail to converge at all.
- ▶ It may be trapped in a **local optimum**.
  - ▶ A serious problem for general NLPs.
  - ▶ The starting point matters.
  - ▶ Some algorithms play some tricks to “try” several local optima.
- ▶ It (typically) requires the domain to be **continuous and connected**.
  - ▶ A nonlinear integer program is very hard to solve.
- ▶ We will point out these difficulties.
  - ▶ Remedies are beyond the scope of this course.

# Assumptions

- ▶ In today's lecture, we will introduce two algorithms to solve **unconstrained** NLP.
  - ▶ **Gradient descent.**
  - ▶ **Newton's method.**
- ▶ We will solve

$$\min_{x \in \mathbb{R}^n} f(x)$$

where  $f(\cdot)$  is a **twice-differentiable** function.

- ▶ Our next step is to learn about **gradients and Hessians**, which are the bases of gradient descent and Newton's method.

# Gradients and Hessians

- For a function  $f : \mathbb{R}^n \rightarrow \mathbb{R}$ , collecting its first- and second-order partial derivatives generates its **gradient** and **Hessian**:

## Definition 1 (Gradients and Hessians)

*For a multi-variate twice differentiable function  $f : \mathbb{R}^n \rightarrow \mathbb{R}$ , its gradient and Hessian are*

$$\nabla f(x) = \begin{bmatrix} \frac{\partial f(x)}{\partial x_1} \\ \vdots \\ \frac{\partial f(x)}{\partial x_n} \end{bmatrix} \quad \text{and} \quad \nabla^2 f(x) = \begin{bmatrix} \frac{\partial^2 f(x)}{\partial x_1^2} & \frac{\partial^2 f(x)}{\partial x_1 \partial x_2} & \cdots \\ \frac{\partial^2 f(x)}{\partial x_2 \partial x_1} & \ddots & \\ \vdots & & \frac{\partial^2 f(x)}{\partial x_n^2} \end{bmatrix}.$$

- In this course, all Hessians are **symmetric**.

## Example

- For  $f(x_1, x_2, x_3) = x_1^2 + x_2x_3 + x_3^3$ , the gradient is

$$\nabla f(x) = \begin{bmatrix} \frac{\partial f(x)}{\partial x_1} \\ \frac{\partial f(x)}{\partial x_2} \\ \frac{\partial f(x)}{\partial x_3} \end{bmatrix} = \begin{bmatrix} 2x_1 \\ x_3 \\ x_2 + 3x_3^2 \end{bmatrix}.$$

- The Hessian is

$$\nabla^2 f(x) = \begin{bmatrix} \frac{\partial^2 f(x)}{\partial x_1^2} & \frac{\partial^2 f(x)}{\partial x_1 \partial x_2} & \frac{\partial^2 f(x)}{\partial x_1 \partial x_3} \\ \frac{\partial^2 f(x)}{\partial x_2 \partial x_1} & \frac{\partial^2 f(x)}{\partial x_2^2} & \frac{\partial^2 f(x)}{\partial x_2 \partial x_3} \\ \frac{\partial^2 f(x)}{\partial x_3 \partial x_1} & \frac{\partial^2 f(x)}{\partial x_3 \partial x_2} & \frac{\partial^2 f(x)}{\partial x_3^2} \end{bmatrix} = \begin{bmatrix} 2 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 6x_3 \end{bmatrix}.$$

- What are  $\nabla f(3, 2, 1)$  and  $\nabla^2 f(3, 2, 1)$ ?

# Road map

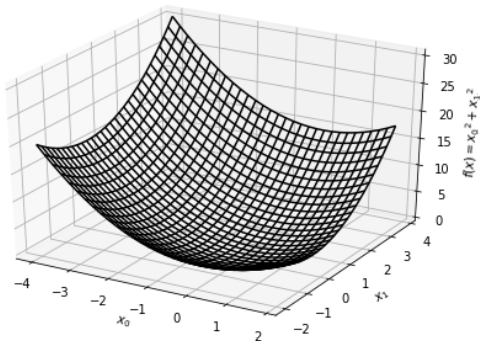
- ▶ Introduction.
- ▶ **Gradient descent.**
- ▶ Newton's method.

# Gradient descent

- ▶ We first introduce the **gradient descent** method.
- ▶ Given a current solution  $x \in \mathbb{R}^n$ , consider its gradient

$$\nabla f(x) = \begin{bmatrix} \frac{\partial f(x)}{\partial x_1} \\ \vdots \\ \frac{\partial f(x)}{\partial x_n} \end{bmatrix}.$$

- ▶ The gradient is an  $n$ -dimensional vector. We may try to “improve” our current solution by moving along this direction.



# Gradient is an increasing direction

- Is the gradient an improving direction?

## Proposition 1

*For a twice-differentiable function  $f(x)$ , its gradient  $\nabla f(x)$  is an increasing direction, i.e.,  $f(x + a\nabla f(x)) > f(x)$  for all  $a > 0$  that is small enough.*

*Proof.* Recall that

$$\lim_{a \rightarrow 0} \frac{f(x + a\nabla f(x)) - f(x)}{a} = \nabla f(x)^T \nabla f(x).$$

Therefore, we have  $\lim_{a \rightarrow 0} \frac{f(x + a\nabla f(x)) - f(x)}{a} = \nabla f(x)^T \nabla f(x) > 0$ , which means that if  $a$  is small enough,  $f(x + a\nabla f(x))$  is greater than  $f(x)$ .  $\square$

- In fact the gradient is the **fastest increasing direction**.

## Gradient is an increasing direction

- ▶ Given that the gradient is an increasing direction, we should move along its opposite direction (for a minimization problem).
- ▶ Therefore, given a current solution  $x$ :
  - ▶ In each iteration we update it to

$$x - a \nabla f(x)$$

for some value  $a > 0$ .  $a$  is called the **step size**.

- ▶ We stop when the gradient of a current solution is 0.
- ▶ Question: How to choose an appropriate value of  $a$ ?
- ▶ Before we answer this question, let's see an example.

# A bad step size can be very bad

- ▶ Let's solve

$$\min_{x \in \mathbb{R}^2} f(x) = x_1^2 + x_2^2.$$

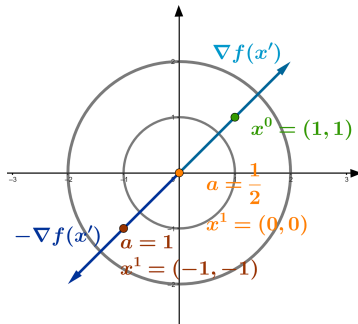
- ▶ Suppose we start at  $x^0 = (1, 1)$ .

- ▶ The gradient in general is  $\nabla f(x) = (2x_1, 2x_2)$ .
- ▶ The gradient at  $x^0$  is  $\nabla f(x^0) = (2, 2)$ .

- ▶ If we set  $a = \frac{1}{2}$ , we will move from  $x^0$  to  $x^1 = (1, 1) - \frac{1}{2}(2, 2) = (0, 0)$ . Optimal!

- ▶ If we set  $a = 1$ , we will move to  $x^1 = (1, 1) - (2, 2) = (-1, -1)$ .

- ▶ The gradient at  $x^1$  is  $\nabla f(x^1) = (-2, -2)$ .
- ▶ We move to  $x^2 = (-1, -1) - (-2, -2) = (1, 1)$ .
- ▶ The algorithm does not converge.



# Maximizing the improvement

- ▶ How to choose a step size?
- ▶ We may instead look for the **largest improvement**.
  - ▶ Along our improving direction  $-\nabla f(x)$ , we solve

$$\min_{a \geq 0} f(x - a \nabla f(x))$$

to see how far we should go to reach the lowest point along this direction.

- ▶ We now may describe our **gradient descent algorithm**.
- ▶ Step 0: Choose a starting point  $x^0$  and a precision parameter  $\epsilon > 0$ .
- ▶ Step  $k + 1$ :
  - ▶ Find  $\nabla f(x^k)$ .
  - ▶ Solve  $a_k = \operatorname{argmin}_{a \geq 0} f(x^k - a \nabla f(x^k))$ .
  - ▶ Update the current solution to  $x^{k+1} = x^k - a_k \nabla f(x^k)$ ,
  - ▶ If  $\|\nabla f(x^{k+1})\| < \epsilon$ , stop; otherwise let  $k$  become  $k + 1$  and continue.<sup>1</sup>

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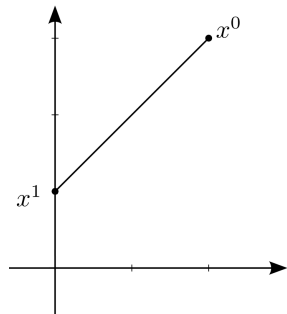
<sup>1</sup>For  $x \in \mathbb{R}^n$ ,  $\|x\| = \sqrt{x_1^2 + \cdots + x_n^2}$ .

## Example 1

- ▶ Let's solve  $\min f(x) = 4x_1^2 - 4x_1x_2 + 2x_2^2$ .
  - ▶ The optimal solution is  $x^* = (0, 0)$ .
  - ▶ We have  $\nabla f(x) = (8x_1 - 4x_2, -4x_1 + 4x_2)$
- ▶ Step 0:  $x^0 = (2, 3)$ .  $f(x^0) = 10$ .
- ▶ Step 1:
  - ▶  $\nabla f(x^0) = (4, 4)$ .
  - ▶  $a_0 = \operatorname{argmin}_{a \geq 0} f(x^0 - a \nabla f(x^0))$ , where

$$\begin{aligned} f(x^0 - a \nabla f(x^0)) &= f(2 - 4a, 3 - 4a) \\ &= 32a^2 - 32a + 10. \end{aligned}$$

- It follows that  $a_0 = \frac{1}{2}$ .
- ▶  $x^1 = x^0 - a_0 \nabla f(x^0) = (2, 3) - \frac{1}{2}(4, 4) = (0, 1)$ .  
Note that  $f(x^1) = 2$ .
  - ▶  $\|\nabla f(x^1)\| = \|(-4, 4)\| = 4\sqrt{2}$ .



## Example 1

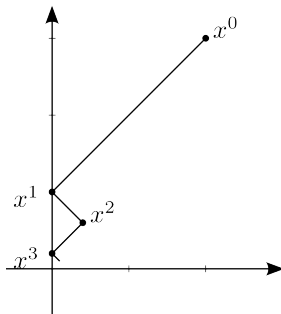
► Step 2:

- $\nabla f(x^1) = (-4, 4)$ .
- $a_1 = \operatorname{argmin}_{a \geq 0} f(x^1 - a \nabla f(x^1))$ , where

$$\begin{aligned} f(x^1 - a \nabla f(x^1)) &= f(0 + 4a, 1 - 4a) \\ &= 160a^2 - 32a + 2. \end{aligned}$$

It follows that  $a_1 = \frac{1}{10}$ .

- $x^2 = x^1 - a_1 \nabla f(x^1) = (0, 1) - \frac{1}{10}(-4, 4) = (\frac{2}{5}, \frac{3}{5})$ . Note that  $f(x^2) = \frac{2}{5}$ .
- $\|\nabla f(x^2)\| = \|(\frac{4}{5}, \frac{4}{5})\| = \frac{4\sqrt{2}}{5}$ .



## Example 2

- ▶ Let's solve  $\min_{x \in \mathbb{R}^2} f(x) = x_1^2 - 2x_1x_2 + 2x_2^2 + 2x_1$ .
  - ▶ We have  $\nabla f(x_1, x_2) = \begin{bmatrix} 2x_1 - 2x_2 + 2 \\ -2x_1 + 4x_2 \end{bmatrix}$ .
  - ▶ We are searching for a point  $x^*$  that satisfies  $\nabla f(x^*) = 0$ . This implies that  $(x_1^*, x_2^*) = (-2, -1)$ .
- ▶ Step 0:  $x^0 = (0, 0)$ .
- ▶ Step 1:
  - ▶ We have

$$\nabla f(x_1^0, x_2^0) = \begin{bmatrix} 2 \\ 0 \end{bmatrix}.$$

- ▶ We may obtain an optimal value for  $a_0$  by solving

$$a_0 = \operatorname{argmin}_{a \geq 0} f(x^0 - a \nabla f(x^0)) = 4a^2 - 4a,$$

it follows that  $a_0 = \frac{1}{2}$ . Therefore,  $x^1 = (0, 0) - \frac{1}{2}(2, 0) = (-1, 0)$ .

## Example 2

► Step 2:

► We have

$$\nabla f(x_1^1, x_2^1) = \begin{bmatrix} 0 \\ 2 \end{bmatrix}.$$

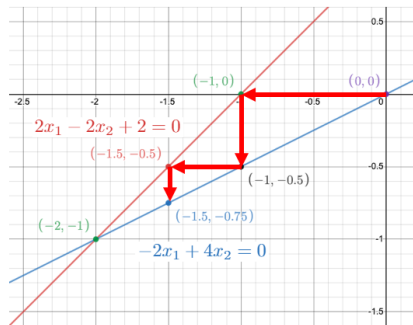
► We may obtain an optimal value for  $a_1$  by solving

$$a_1 = \operatorname{argmin}_{a \geq 0} f(x^1 - a \nabla f(x^1)) = 8a^2 - 4a - 1,$$

it follows that  $a_1 = \frac{1}{4}$ . Therefore,  $x^2 = (-1, 0) - \frac{1}{4}(0, 2) = (-1, -\frac{1}{2})$ .

## Example 2

- ▶ By depicting the search route from  $x^0$  to  $x^1$  to  $x^2$ , and obtaining optimal  $x^*$  by the FOC, we know that the algorithm search only one direction once a time.
- ▶ Thus, we can predict  $x^3$  is on  $2x_1 - 2x_2 + 2 = 0$  with  $x_2 = -\frac{1}{2}$ , so  $x^3 = (-\frac{3}{2}, -\frac{1}{2})$ . And  $x^4 = (-\frac{3}{2}, -\frac{3}{4})$ . By doing more iterations, you can get the optimal solution  $x^* = (-2, -1)$ .



# Road map

- ▶ Introduction.
- ▶ Gradient descent.
- ▶ **Newton's method.**

# Newton's method

- ▶ The gradient descent method is a **first-order** method.
  - ▶ It relies on the gradient to improve the solution.
- ▶ A first-order method is intuitive, but sometimes too slow.
- ▶ A **second-order** method relies on the Hessian to update a solution.
- ▶ We will introduce one second-order method: **Newton's method**.
- ▶ Let's start from Newton's method for solving a **nonlinear equation**.

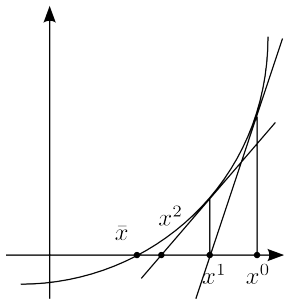
# Newton's method for a nonlinear equation

- ▶ Let  $f : \mathbb{R} \rightarrow \mathbb{R}$  be differentiable. We want to find  $\bar{x}$  satisfying  $f(\bar{x}) = 0$ .
- ▶ For any  $x^k$ , let

$$f_L(x) = f(x^k) + f'(x^k)(x - x^k)$$

be the **linear approximation** of  $f$  at  $x^k$ .

- ▶ This is the **tangent line** of  $f$  at  $x^k$  or the first-order **Taylor expansion** of  $f$  at  $x^k$ .
- ▶ We move from  $x^k$  to  $x^{k+1}$  by setting
$$f_L(x^{k+1}) = f(x^k) + f'(x^k)(x^{k+1} - x^k) = 0.$$
- ▶ We will keep iterating until  $|f(x^k)| < \epsilon$  or  $|x^{k+1} - x^k| < \epsilon$  for some predetermined  $\epsilon > 0$ .



## Newton's method for single-variate NLPs

- ▶ Let  $f$  be twice differentiable. We want to find  $\bar{x}$  satisfying  $f'(\bar{x}) = 0$ .
- ▶ For any  $x^k$ , let

$$f'_L(x) = f'(x^k) + f''(x^k)(x - x^k)$$

be the **linear approximation** of  $f'$  at  $x^k$ .

- ▶ To approach  $\bar{x}$ , we move from  $x^k$  to  $x^{k+1}$  by setting

$$f'_L(x^{k+1}) = f'(x^k) + f''(x^k)(x^{k+1} - x^k) = 0.$$

- ▶ We will keep iterating until  $|f'(x^k)| < \epsilon$  or  $|x^{k+1} - x^k| < \epsilon$  for some predetermined  $\epsilon > 0$ .
- ▶ Note that  $f'(\bar{x})$  does not guarantee a global minimum.
  - ▶ That is why showing  $f$  is convex is useful!

## Another interpretation

- ▶ Let  $f$  be twice differentiable. We want to find  $\bar{x}$  satisfying  $f'(\bar{x}) = 0$ .
- ▶ For any  $x^k$ , let

$$f_Q(x) = f(x^k) + f'(x^k)(x - x^k) + \frac{1}{2}f''(x^k)(x - x^k)^2$$

be the **quadratic approximation** of  $f$  at  $x^k$ .

- ▶ This is the second-order **Taylor expansion** of  $f$  at  $x^k$ .
- ▶ We move from  $x^k$  to  $x^{k+1}$  by moving to the **global minimum** of the quadratic approximation, i.e.,

$$x^{k+1} = \operatorname{argmin}_{x \in \mathbb{R}} f(x^k) + f'(x^k)(x - x^k) + \frac{1}{2}f''(x^k)(x - x^k)^2,$$

- ▶ Differentiating the above objective function with respect to  $x$ , we have

$$f'(x^k) + f''(x^k)(x^{k+1} - x^k) = 0 \quad \Leftrightarrow \quad x^{k+1} = x^k - \frac{f'(x^k)}{f''(x^k)}.$$

## Example: the NLP

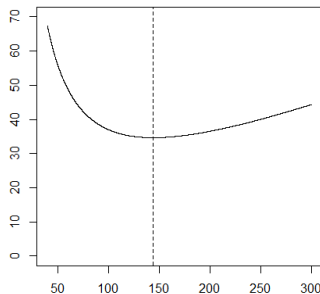
► Let

$$f = \frac{KD}{x} + \frac{hx}{2},$$

where  $K = 5$ ,  $D = 500$ , and  $h = 0.24$ .

► The global minimum is

$$x^* = \sqrt{\frac{2KD}{h}} \approx 144.34.$$



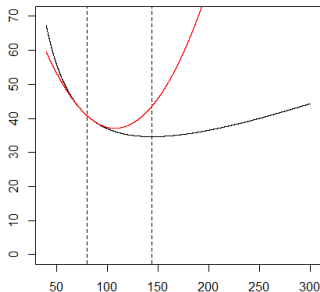
## Example: quadratic approximation

- At any  $x^k$ , the quadratic approximation is

$$\begin{aligned} & f(x^k) + f'(x^k)(x - x^k) + \frac{1}{2}f''(x^k)(x - x^k)^2 \\ &= \left( \frac{KD}{x^k} + \frac{hx^k}{2} \right) + \left( \frac{-KD}{(x^k)^2} + \frac{h}{2} \right)(x - x^k) \\ &\quad + \frac{1}{2} \left( \frac{2KD}{(x^k)^3} \right) (x - x^k)^2. \end{aligned}$$

- E.g., at  $x^0 = 80$ , it is (approximately)

$$40.85 - 0.27(x - 80) + 0.0049(x - 80)^2.$$



## Example: one iteration

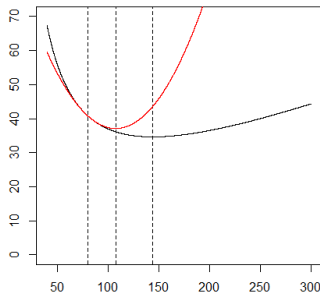
- ▶ At any  $x^k$ , the quadratic approximation may be obtained.
- ▶ Its global minimum  $x^{k+1}$  satisfies

$$\left( \frac{-KD}{(x^k)^2} + \frac{h}{2} \right) + \left( \frac{2KD}{(x^k)^3} \right) (x^{k+1} - x^k) = 0.$$

- ▶ E.g., at  $x^0 = 80$ , we have

$$-0.27 + 0.0098(x^1 - 80) = 0,$$

i.e.,  $x^1 \approx 107.71$ .

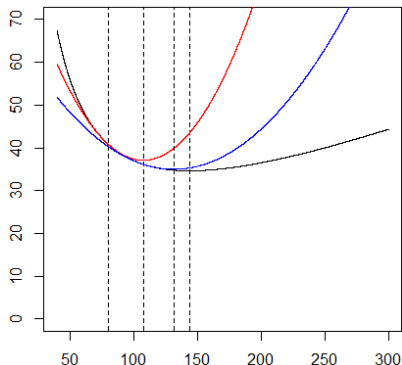


## Example: one more iteration

- ▶ Note that from  $x^k$  we may simply move to

$$x^{k+1} = x^k - \frac{\frac{-KD}{(x^k)^2} + \frac{h}{2}}{\frac{2KD}{(x^k)^3}}.$$

- ▶ From  $x^1 = 107.71$ , we will move to  $x^2 = 131.58$ .
- ▶ We get closer to  $x^* = 144.34$ .



## Newton's method for multi-variate NLPs

- ▶ Let  $f : \mathbb{R}^n \rightarrow \mathbb{R}$  be twice differentiable.
- ▶ For any  $x^k$ , let

$$f_Q(x) = f(x^k) + \nabla f(x^k)^T(x - x^k) + \frac{1}{2}(x - x^k)^T \nabla^2 f(x^k)(x - x^k)$$

be the quadratic approximation of  $f$  at  $x^k$ .

- ▶ Note that we use the **Hessian**  $\nabla^2 f(x^k)$ .
- ▶ We move from  $x^k$  to  $x^{k+1}$  by moving to the global minimum of the quadratic approximation:

$$\nabla f(x^k) + \nabla^2 f(x^k)(x^{k+1} - x^k) = 0,$$

i.e.,

$$x^{k+1} = x^k - \left[ \nabla^2 f(x^k) \right]^{-1} \nabla f(x^k).$$

## Example

- ▶ Let's minimize  $f(x) = x_1^4 + 2x_1^2x_2^2 + x_2^4$ .

- ▶ The optimal solution is  $x^* = (0, 0)$ .

- ▶  $\nabla f(x) = \begin{bmatrix} 4x_1^3 + 4x_1x_2^2 \\ 4x_1^2x_2 + 4x_2^3 \end{bmatrix}$  and  $\nabla^2 f(x) = \begin{bmatrix} 12x_1^2 + 4x_2^2 & 8x_1x_2 \\ 8x_1x_2 & 12x_2^2 + 4x_1^2 \end{bmatrix}$ .

- ▶ Suppose that  $x^0 = (b, b)$  for some  $b > 0$ .

- ▶ We have  $\nabla f(x^0) = \begin{bmatrix} 8b^3 \\ 8b^3 \end{bmatrix}$  and  $\nabla^2 f(x^0) = \begin{bmatrix} 16b^2 & 8b^2 \\ 8b^2 & 16b^2 \end{bmatrix}$ .

- ▶ Therefore, we have

$$\begin{aligned} x^1 &= x^0 - \left[ \nabla^2 f(x^0) \right]^{-1} \nabla f(x^0) \\ &= \begin{bmatrix} b \\ b \end{bmatrix} - \frac{1}{192b^2} \begin{bmatrix} 16 & -8 \\ -8 & 16 \end{bmatrix} \begin{bmatrix} 8b^3 \\ 8b^3 \end{bmatrix} = \begin{bmatrix} \frac{2}{3}b \\ \frac{2}{3}b \end{bmatrix}. \end{aligned}$$

- ▶ In fact, we have  $x^k = \left( \left(\frac{2}{3}\right)^k b, \left(\frac{2}{3}\right)^k b \right)$ .

## Remarks

- ▶ For Newton's method:
  - ▶ Newton's method does not have the step size issue.
  - ▶ It in many cases is faster.
  - ▶ For a quadratic function, Newton's method finds an optimal solution in one iteration.
  - ▶ It may fail to converge for some functions.
- ▶ More issues in general:
  - ▶ Convergence guarantee.
  - ▶ Convergence speed.
  - ▶ Non-differentiable functions.
  - ▶ Constrained optimization.